

Arrange Flowers in a Vase SOP

One episode takes three flowers from a tray, lays them out on a cutting board, cuts every stem to length with a blade, strips the low leaves off each one, stands all three in a vase tallest to shortest, pours water into the vase, and wipes the drips off the table and the vase.

The order never changes: lay out all three, cut all three, strip all three, then set them in the vase. Inside every one of those, the tall flower comes first, then the middle flower, then the short flower. The water goes in only once all three stand in the vase, and the drips are wiped last.

The left gripper carries every flower and holds it flat against the cutting board while it is cut and stripped, then stands it in the vase. The right gripper works the blade, pulls the leaves, clears the waste, pours the water, and wipes. Neither gripper crosses to the other side of the table.

Every flower is cut and stripped lying flat on the cutting board, never held up in the air. A flower is raised upright only at the moment it goes into the vase, and it is never held by its bloom. Every stem is cut with the blade drawn toward the front edge along the trim line, stroke after stroke on that same line until the stem parts. The number of strokes is not fixed and is never counted. The vase is never gripped, steadied, or nudged at any point in the episode.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the flower tray at the back left, the vase at the center back, the cutting board at the center front, the blade rest right of the cutting board, the pitcher at the back right, the cloth rest at the front right, and the waste tray at the front right corner.
3. The cutting board is visible from above, so the trim line and every cut can be seen.
4. The vase is visible from the side, so the water rising inside it can be seen.
5. Both arms are at home with grippers open.
6. The table is clean, dry, and bare apart from the flower tray, the vase, the cutting board, the blade, the pitcher, the cloth, and the waste tray.
7. The left arm reaches the flower tray, the cutting board, and the vase. The right arm reaches the blade rest, the cutting board, the vase, the pitcher spot, the cloth rest, and the waste

tray.

Materials checklist

1. Three **flowers** lie side by side in the flower tray at the back left, blooms toward the left edge and stem ends toward the right. Their stem lengths are clearly different: the **tall flower**, the **middle flower**, and the **short flower**. Anyone looking at the tray can tell which is which.
2. Every flower has leaves on the lower half of its stem, and a stem firm enough to lie straight rather than curl on the board.
3. One **cutting board** sits at the center front, clean and dry, with the **trim line** marked across it a short way in from its right edge. The board is wide enough to hold all three flowers side by side without their blooms touching.
4. One **blade** lies flat on the blade rest right of the cutting board, edge down and handle toward the front edge. Its handle runs along its back so one gripper can close on it, and it is sharp enough to part a sound stem cleanly rather than crush or tear it.
5. One **vase** stands at the center back, empty and dry inside and out. Its base is wide and heavy enough that it stands with three flowers in it and takes a pour without being held, and its wall is clear enough to show the water level from the side.
6. The vase rim reaches no higher than the middle of the shortest stem once that stem is cut.
7. One **pitcher** stands on the table at the back right, handle toward the front edge, holding enough water to fill the vase to the fill level with water to spare.
8. One folded dry **cloth** per episode lies on the cloth rest at the front right. A cloth is used for one episode and thrown away.
9. One **waste tray**, empty, at the front right corner.

Workspace layout

- **Flower tray:** back left, the three flowers lying flat, blooms left and stem ends right.
- **Cutting board:** center front, the working area, carrying the trim line.
- **Vase:** center back, empty, standing.
- **Blade rest:** right of the cutting board, the blade lying flat.
- **Pitcher spot:** back right, the pitcher of water standing on the table.
- **Cloth rest:** front right, the folded cloth.
- **Waste tray:** front right corner, holding the cut ends, the stripped leaves, and the used cloth.

The only thing the left arm carries is a flower, and it moves along the left and the center. Everything the right arm carries is staged on the right. No carried thing crosses the midline.

Vocabulary

- **Arm assignment:** the left gripper carries every flower, holds it flat against the cutting board while the right gripper works on it, and stands it in the vase. The right gripper takes the blade and cuts, pulls the low leaves, clears the cut ends and the leaves to the waste tray, pours the water, and wipes. Neither gripper takes the other's zones.
- **Bloom:** the flower head at the top of the stem. It is never gripped and never touches the blade, the waste tray, or the cloth.
- **Stem:** the length of the flower below the bloom. The **stem end** is the bottom of it, and the **grasp point** is around the middle of the stem, above the low leaves and clear of the trim line.
- **Table hold:** the left gripper closed on a stem and holding it flat against the cutting board while the right gripper works on it. The hold sits on the bloom side of the trim line, clear of the blade path. The stem never leaves the board while it is held.
- **Stem end line-up:** the three stems lie with their stem ends level with one another at the right edge of the cutting board, so every stem is cut the same length shorter and the three flowers keep their order of height.
- **Trim line:** the line marked across the cutting board. Every stroke of the blade lands on it.
- **Slice stroke:** the edge set square across the stem on the trim line and drawn once toward the front edge while it presses down into the stem, then lifted clear of the stem before it goes back for the next stroke. The blade cuts on the draw only. It never saws back and forth, never cuts on the way back, and never travels along the stem away from the trim line.
- **Cutting a stem:** slice strokes repeated in the same cut on the trim line until the stem parts. How many strokes a stem takes is not fixed and is never counted, because no single stroke parts a flower stem. A thick stem takes more than a thin one. What matters is that every stroke lands in the same cut, the table hold stays on the whole time, and the stem parts rather than crushes.
- **Parted stem:** the stem separated at the trim line, the cut end lying free between the trim line and the right edge of the board, its face square rather than crushed or torn.
- **Stalled cut:** the strokes stop biting and the stem flattens, crushes, or tears at the trim line instead of parting. A stalled cut ends the episode and is logged as a blade fault.
- **Cut end:** the short piece of stem left between the trim line and the right edge of the board once the stem parts.
- **Low leaf:** any leaf on the lower half of the stem, counting from the cut end up to the bloom. Every low leaf comes off. Nothing above the halfway point is touched.
- **Vase mouth:** the open top of the vase. It holds three places: the **back of the mouth** farthest from the front edge, the **middle of the mouth**, and the **front of the mouth** nearest the front edge.

- **Stepped arrangement:** the tall flower at the back of the mouth, the middle flower in the middle, and the short flower at the front, so the blooms step down toward the front edge. No stem crosses another and no bloom leans on another.
- **Fill level:** the water standing halfway up the vase, seen from the side. The pour stops the moment the water reaches it.
- **Unheld vase:** from setup to the end of the episode the vase is never gripped, steadied, or nudged. It stands on its own weight. If it is knocked out of place, end the episode and put it right during the reset, not while recording.
- **Drip:** water on the tabletop or running down the outside of the vase.
- **Cloth:** the folded cloth used to wipe once, then thrown away. Its **cloth face** is the flat side that touches the table.
- **Wipe pass:** one continuous draw of the cloth face from left to right across the table, parallel to the front edge.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Release point:** over the cutting board, over the vase mouth, over the blade rest, over the pitcher spot, or over the waste tray. A gripper opens nowhere else.

Steps

Steps 1 to 3 prepare the three flowers on the cutting board. Steps 4 and 5 finish the vase. Step 6 ends the episode.

Step 1: Lay out and trim the three stems

Goal: three trimmed flowers lying on the cutting board, every stem cut on the trim line, and the three cut ends in the waste tray.

1.1 Lay the three flowers on the cutting board

- The **left gripper** closes on the **stem** of the **tall flower** at its **grasp point** in the flower tray, lifts it straight up, carries it level to the **cutting board**, lays it flat with the stem running left to right and the **stem end** at the right edge of the board, and releases.
- Lay it farthest from the front edge, leaving room for two more in front of it.
- Repeat for the **middle flower** and then the **short flower**, laying each one nearer the front edge than the last, stems parallel and stem ends level with one another at the right edge of the board.

Take each flower by the stem only. Never close on the bloom, never lift two flowers in one grasp, and never drag a flower across the table.

Expected state: three flowers lie in a row on the cutting board, tall farthest from the front edge and short nearest it, every stem end level with the others at the right edge of the board, no

bloom touching another.

1.2 Take the blade

- The **right gripper** closes on the **blade** handle on the blade rest and lifts the blade straight up, edge down.

Carry the blade edge down and level. Never turn it in the gripper and never carry it over the vase.

1.3 Cut each stem

- The **left gripper** takes a **table hold** on the tall flower on the bloom side of the trim line.
- The **right gripper** brings the blade over the stem where it crosses the **trim line**, square across the stem, and takes a **slice stroke** toward the front edge.
- Lift the edge clear of the stem, set it back into the same cut, and take another slice stroke.
- Repeat slice strokes in that one cut until the stem parts. Do not count them and do not stop at a set number.
- Once the stem has **parted**, lift the blade clear and the **left gripper** releases. That flower is done.
- If the cut **stalls**, end the episode and log a blade fault.
- Repeat for the middle flower, then the short flower.

Every stroke lands in the same cut on the trim line. The table hold stays on from the first stroke until the stem has parted and the blade is clear. Never cut where a gripper is, never cut a stem that is not held flat, never saw the blade back and forth or cut on the way back, and never take another stroke on a stem that has already parted.

1.4 Return the blade and clear the cut ends

- The **right gripper** carries the blade to the **blade rest**, lowers it flat with the edge down and the handle toward the front edge, and releases.
- The **right gripper** then closes on each **cut end** between the trim line and the right edge of the board, lifts it, carries it to the **waste tray**, and releases it over the tray.
- Repeat until no cut end is left on the board.

Check: three flowers lie on the cutting board with every stem cut on the trim line and every cut end square rather than crushed or torn, the three cut ends are in the waste tray, the blade is flat on the blade rest, and nothing has been cut off any stem above the trim line.

Step 2: Strip the low leaves

Goal: no low leaf left on any of the three stems, and every stripped leaf in the waste tray.

- The **left gripper** takes a **table hold** on the tall flower.

- The **right gripper** closes on the lowest **low leaf** where it joins the stem, pulls it away from the stem until it comes off, carries it to the **waste tray**, and releases it over the tray.
- Repeat, working up the stem, until no low leaf is left on the lower half of it, then the **left gripper** releases.
- Repeat for the middle flower, then the short flower.

Pull every leaf away from the bloom, along the stem toward the cut end. Never pull the stem out of the table hold, never take a leaf above the halfway point, and never drop a leaf on the board or the table.

Check: no low leaf is left on the lower half of any of the three stems, no leaf base is left torn onto a stem, every stripped leaf is in the waste tray, and the three flowers still lie in their row with their stem ends level at the right edge of the board.

Step 3: Set the flowers in the vase

Goal: all three flowers standing in the vase as a stepped arrangement.

- The **left gripper** closes on the **tall flower** at its **grasp point**, lifts it clear of the board, brings the stem upright in one smooth motion as it rises, carries it upright to the **vase**, lowers it into the **back of the mouth** until the stem end touches the vase floor, and releases.
- Repeat for the **middle flower** into the **middle of the mouth**, then the **short flower** into the **front of the mouth**.

Bring the stem upright while the flower is clear of the board and the vase, and keep it upright from there to the mouth. Never take a new hold in the air, never lean a stem against the vase rim to guide it, and never lower a flower into a place that already holds one.

Check: three flowers stand in the vase as a stepped arrangement with the blooms stepping down toward the front edge, no stem crosses another, no bloom leans on another, no leaf sits inside the vase or touches the vase rim, the cutting board is clear, and the vase has not moved.

Step 4: Pour water into the vase

Goal: the water standing at the fill level and the pitcher back at its spot.

- The **right gripper** closes on the **pitcher** handle at the pitcher spot and lifts the pitcher straight up.
- Carry it upright to the vase and bring the spout over a gap in the **vase mouth** that is clear of the stems.
- Tilt until the water runs into the vase, and stop the moment the water reaches the **fill level**.
- Bring the pitcher upright over the vase mouth and hold for 2 seconds so it stops dripping.
- Carry it upright to the **pitcher spot**, lower it until it sits flat, and release.

Pour into the mouth only. Never pour onto a bloom or down the outside of the vase, never rest the spout on the vase rim, and never carry the pitcher away from the vase still tilted.

Check: the water stands at the fill level, the pitcher stands upright at its spot, no flower has been pushed out of its place in the mouth, and the vase has not moved.

Step 5: Wipe the drips

Goal: the tabletop and the outside of the vase dry, and the cloth in the waste tray.

- The **right gripper** closes on the folded **cloth** on the cloth rest, carries it to the table in front of the vase, and lowers it until the **cloth face** lies flat on the table.
- Draw the cloth from left to right in one **wipe pass**, keeping the cloth face flat for the whole pass.
- Lift at the end of the pass, return to the left, and start the next wipe pass so it overlaps the one just made, working from the vase base out toward the front edge, until no water is left on the table.
- For a drip running down the outside of the vase, press the cloth face against the vase just enough to touch it and draw it straight down once. Never push the vase sideways.
- Carry the cloth to the **waste tray** and release it over the tray.

Wipe around the vase, never over the vase mouth. The cloth never touches a bloom, a stem, or the water in the vase.

Check: no water is left on the tabletop or on the outside of the vase, the cloth is in the waste tray, the stepped arrangement is unchanged, and the vase has not moved.

Step 6: Confirm the arrangement and end the episode

- Confirm the vase holds three flowers in a stepped arrangement, the water stands at the fill level, the tabletop and the outside of the vase are dry, the flower tray and the cutting board are clear, the blade is flat on the blade rest, the pitcher is upright at its spot, and the waste tray holds the three cut ends, every stripped leaf, and the cloth.
- Return both arms home, clear of the vase and the cutting board, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the three flowers out of the vase and clear them away.
2. Tip the water out of the vase, dry it inside and out, and stand it back at the center back.
3. Empty the waste tray of cut ends, leaves, and the used cloth, wipe it dry, and set it back at the front right corner.
4. Lay three fresh flowers side by side in the flower tray at the back left, blooms toward the left edge and stem ends toward the right, one clearly tall, one clearly middle, one clearly short,

and confirm each has leaves on the lower half of its stem.

5. Wipe the cutting board clean and dry, clear it of stem pieces and leaf scraps, and set it back at the center front with the trim line running across it.
6. Lay the blade flat on the blade rest, edge down and handle toward the front edge, and confirm the edge is clean, dry, and undamaged.
7. Refill the pitcher and stand it back on the table at the back right, handle toward the front edge.
8. Lay one folded dry cloth on the cloth rest at the front right.
9. Wipe any water off the table and confirm the surface is clean and dry. A wet table lets the cutting board and the vase slide.
10. Replace a flower whose stem is bent, split, or too limp to lie straight on the board, and a vase that is chipped, leaks, or rocks on the table.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a stem is cut before all three flowers are on the board, a flower is stripped before all three are cut, a flower goes into the vase before all three are stripped, the water goes in before all three stand in the vase, or the wipe happens before the pour. Inside any of those, the flowers are worked out of order rather than tall, middle, short.
- **SOP rule broken:** Steps 1 to 5, lay out all three, cut all three, strip all three, set all three, then pour, then wipe, taking the flowers tall, middle, short each time.
- **Coaching note:** finish the whole board before you move on, and always work tall to short.

Violation: Arms swapped roles

- **Visible cue:** the right gripper carries a flower from the tray or sets one in the vase, or the left gripper takes the blade, pulls a leaf, clears a cut end, takes the pitcher, or takes the

cloth.

- **SOP rule broken:** Steps 1 to 5, the left gripper carries and holds the flower and the right gripper works the blade, the leaves, the waste, the pour, and the wipe.
- **Coaching note:** each arm stays on its own side. Nothing crosses the midline.

Violation: Flowers laid out wrong

- **Visible cue:** at the end of Step 1.1 a stem end sits short of or past the right edge of the board instead of level with the other two, a stem lies across another, the three are not in tall to short order from the back edge to the front edge, a bloom overlaps another, or a flower is dragged across the table instead of lifted.
- **SOP rule broken:** Step 1.1, three flowers laid flat in a row, stems parallel, every stem end level with the others at the right edge of the board, tall farthest from the front edge and short nearest it.
- **Coaching note:** the stem ends lining up is what makes every cut the same. Line the end up with the other two before you release.

Violation: Stem cut wrong

- **Visible cue:** a stroke lands off the trim line or opens a second cut beside the first, the blade saws back and forth or cuts on the way back, the blade drags along the stem away from the trim line, a stroke is taken while the left gripper is off the stem or while the stem shifts, the blade keeps working a stem that has already parted, a stalled cut is worked on instead of ending the episode, a stem is crushed or torn rather than parted, or a cut end is left on the board.
- **SOP rule broken:** Steps 1.3 and 1.4, slice strokes repeated in one cut on the trim line with the table hold on until the stem parts, and every cut end carried to the waste tray.
- **Coaching note:** hold first, line the edge up on the trim line, then draw toward you and lift clear before you come back. Keep every stroke in the same cut and stop the moment the stem parts.

Violation: Blade handled wrong

- **Visible cue:** the blade is carried edge up or tilted, turned in the gripper, carried over the vase, swung across the board between cuts, set down anywhere other than the blade rest, or left off the blade rest once the last stem is cut.
- **SOP rule broken:** Steps 1.2 and 1.4, the blade is carried edge down and level, moves only along the trim line while it cuts, and goes back flat on the blade rest.
- **Coaching note:** edge down, draw along the line, lift clear, and straight back to the rest once the last stem has parted.

Violation: Low leaves not stripped

- **Visible cue:** at the end of Step 2 a leaf is still on the lower half of a stem, a leaf above the halfway point is pulled off, a leaf is torn so its base stays on the stem, a stem is pulled out of the table hold, or a stripped leaf is left on the board or the table instead of going to the waste tray.
- **SOP rule broken:** Step 2, every low leaf comes off, nothing above the halfway point is touched, and every leaf is carried to the waste tray.
- **Coaching note:** start at the lowest leaf and work up. Pull away from the bloom, then carry it to the tray.

Violation: Flower raised or carried wrong

- **Visible cue:** a flower is gripped by the bloom or the leaves, held up in the air while it is cut or stripped, laid back down on the table between the board and the vase, given a new hold in the air, or leaned against the vase rim to guide it into the mouth.
- **SOP rule broken:** Steps 1.1, 1.3, 2, and 3, the flower is taken by the stem, worked lying flat on the board, and brought upright only on its way into the vase.
- **Coaching note:** stem only. Work it on the board, stand it up once, and take it straight in.

Violation: Vase gripped or knocked out of place

- **Visible cue:** a gripper closes on the vase, steadies it, or nudges it, or the vase is struck and shifts, rocks, or turns out of its place at the center back.
- **SOP rule broken:** Steps 3 to 5, the vase stands on its own weight and is never gripped, steadied, or nudged.
- **Coaching note:** work around the vase. If it moves, end the episode and set it right during the reset.

Violation: Pour wrong

- **Visible cue:** the water goes short of or past the fill level, water lands on a bloom, on the outside of the vase, or on the table, the spout rests on the vase rim, the stream is broken and restarted, the pitcher is carried away still tilted, or a flower is pushed out of its place by the pour.
- **SOP rule broken:** Step 4, one pour into the mouth clear of the stems, stopping the moment the water reaches the fill level, with the pitcher upright before it leaves the vase.
- **Coaching note:** watch the water rising, not the pitcher. Bring it upright over the mouth and hold before you carry it back.

Violation: Wipe skipped, misdirected, or drips left

- **Visible cue:** Step 5 is skipped, water is still on the table or on the outside of the vase at the end of it, the wipe passes run right to left, front to back, or on the diagonal, a strip of table is left untouched between two passes, or the cloth passes over the vase mouth or touches a bloom, a stem, or the water.

- **SOP rule broken:** Step 5, left to right wipe passes working from the vase base out to the front edge, each overlapping the one just made, until the table and the vase outside are dry.
- **Coaching note:** keep every pass parallel to the front edge and overlap the last one. Wipe around the vase, never across the mouth.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** a flower, the blade, the pitcher, or the cloth falls to the table or the floor short of its spot, the cutting board, the flower tray, or the waste tray is struck and shifts or tips, the pitcher spills, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 5, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a flower still in the flower tray or on the cutting board, the water short of or past the fill level, water on the table or on the vase, a cut end or a leaf outside the waste tray, the blade off its rest, the pitcher away from its spot, the cloth on the table, an arm away from home, or an arm still over the vase.
- **SOP rule broken:** Step 6, confirm the vase, the water, the dry table, and the clear zones; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective flower:** a stem bent, split, or too limp to lie straight on the board under a correct table hold, or a bloom that falls off the stem. Replace it before the next episode.
- **Blade fault:** the cut stalls, so correct slice strokes stop biting and the edge crushes or tears a sound stem instead of parting it, the edge is chipped, or the handle will not sit in one grasp. Replace it before the next episode.
- **Defective vase:** chipped, leaking, rocking on the table, or so light that it shifts under a correct pour. Replace it before the next episode.
- **Pitcher fault:** it will not pour a controllable stream, or it runs dry before the water reaches the fill level. Refill or replace it before the next episode.
- **Defective cloth:** already wet, torn, or shedding fibre onto the table. Replace it before the next episode.

- **Cutting board slides** under a correct slice stroke, or the flower tray or waste tray slides under a correct pick, because the table is wet or slick. Wipe and dry the table before the next episode.

Annotation subtasks (from SOP)

1. Lay the three flowers in a row on the cutting board with their stem ends level
2. Cut the three stems on the trim line with the blade
3. Clear the three cut ends to the waste tray
4. Strip the low leaves off the three stems
5. Set the tall flower at the back of the vase mouth
6. Set the middle flower in the middle of the vase mouth
7. Set the short flower at the front of the vase mouth
8. Pour water into the vase to the fill level
9. Wipe the drips off the table and the vase
10. Confirm the arrangement and end the episode